```
SD.open(filename, mode)
```

The `filename` parameter is a character string for the name of the file that we want to access. This name can also include the directory structure on the card using forward slashes, `/`, to separate directories. If the file does not exist, this function will create it, although the directory must already exist beforehand. This function only works with short file names from the days of yore in an 8.3 format. This means an 8-character name with a 3-character file extension separated by a period, as in `datafile.txt`.

There are two possible modes for this function: `FILE_READ` and `FILE_WRITE`. The default mode for the function when no mode is specified is `FILE_READ`. This mode will open the specified file at the beginning of the file to be read by the `read()` function explained later. The `FILE-WRITE` mode is used to write information to the file beginning at the end of the file. In other words, we read from the beginning of the file and we write to the end of a file. In our example code it is as follows:

```
file = SD.open("data.txt", FILE_WRITE);
```

We opened the file `data.txt` and set it to a `FILE_WRITE` mode so that we could write data to the file and then attributed this to the instance `file` that we created earlier. In our next few functions, the instance name `file` will be used to tell the functions what file to use.

■ **Note** Keep in mind that only one file can be opened at a time.

---

close()

When we are done with the file that we have open, or if we need to open a new file, we must first close the current file using the `close()` function to save the data that has been written to the SD card. The syntax for this function follows:

```
name.close();
```

The specified name is the name of the `File` instance that we have been working with, as in `file.close();` from our example code earlier, and has no further parameters and does not return any data. This function has been called after the data has been written to the file to save the file before moving on.

write()

Used in the sketch to write a data value to the SD card, the `write()` function has the following syntax:

```
name.write(data)
```

The name is the instance of the `File` class created at the beginning of the code and data can be a numerical value of the `byte`, `char`, or `string` data type. In our example we used the following code:

```
byte value = map(analogRead(sensor), sensorMin, sensorMax, 0, 255);
file.write(value);
```

These two lines read a value from an analog input, map those values to a range of 0–255, and assign that value to the byte named `value`. This value is then written to our file.